

Problem 3

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Problem 1 Circle the correct answer in each part:

(a) Consider the limit $\lim_{x \rightarrow 0} (\cos(x))^{\sin(x)}$.

(i) Evaluate the limit.

i. the limit DNE

ii. e

iii. 1

iv. ∞

v. $-\infty$

vi. 0

vii. none of the previous answers is correct

Explanation. The correct choice is (iii).

Evaluation of limit:

$$\lim_{x \rightarrow 0} \underbrace{(\cos(x))^{\sin(x)}}_{\text{form } 1^0} = 1$$

(ii) What Limit Law, rule or technique did you use to find this limit?

i. The Squeeze Theorem;

ii. L'Hôpital's Rule;

iii. The Product Law;

iv. evaluated the function at $x = 0$, since the function is continuous at $x = 0$;

v. none of the previous answers is correct

Explanation. The correct choice is (iv).

(b) Evaluate the limit $\lim_{x \rightarrow 4^-} \frac{\ln(x)}{x - 4}$.

(i) the limit DNE

(ii) e

(iii) 1

(iv) ∞

(v) $-\infty$

Problem 3

- (vi) 0
- (vii) *none of the previous answers is correct*

Explanation. The correct choice is (v).

Evaluation of limit: $\lim_{x \rightarrow 4^-} \frac{\ln(x)}{x - 4} = -\infty$,

since the limit is of the form $\frac{\#}{0}$, and since the numerator is positive and the denominator negative.

(c) Evaluate the limit $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x - 4}$.

- (i) *the limit DNE*
- (ii) e
- (iii) 1
- (iv) ∞
- (v) $-\infty$
- (vi) 0
- (vii) *none of the previous answers is correct*

Explanation. The correct choice is (vi).

Evaluation of limit:

$$\begin{aligned} \lim_{x \rightarrow \infty} \underbrace{\frac{\ln(x)}{x - 4}}_{\text{form } \infty/\infty} &\stackrel{L.H.}{=} \lim_{x \rightarrow \infty} \frac{1/x}{1} \\ &= 0 \end{aligned}$$

(d) Consider the limit

$$\lim_{h \rightarrow 0} \frac{(2 + h)^3 - 8}{h} = f'(2).$$

Determine the function f .

- (i) *such a function DNE;*
- (ii) $f(x) = x^3$;
- (iii) $f(x) = (2 + x)^3$;
- (iv) $f(x) = \frac{(2 + x)^3}{x}$;
- (v) *none of the previous answers is correct*

Explanation. The correct choice is (ii).

Problem 3

(e) Consider the limit: $\lim_{x \rightarrow 0^+} \left(\frac{\sin(x)}{x} \right)^{|\ln(x)|}$.

Determine the form of this limit.

- (i) $\frac{0}{0}$;
- (ii) $\frac{\infty}{\infty}$;
- (iii) 1^0 ;
- (iv) 0^0 ;
- (v) 1^∞ ;
- (vi) ∞^∞ ;
- (vii) none of the previous answers is correct

Explanation. The correct choice is (v).